Contents

0.1 Construction of Hamiltonian . . . 2
0.1.1 Construction of the Hamiltonian . . . 2
0.1.2 Continuity equation and probability current . . . 3
0.2 0-0 and π-π States . . . 5
0.2.1 Pedestrian method . . . 5
0.2.2 Pedro's method . . . 8
0.3 $\pi/2$-$\pi/2$ and $-\pi/2$ - $-\pi/2$ States . . . 9
0.3.1 Pedestrian Method . . . 9
0.3.2 Pedro's Method . . . 11
0.3.3 Chern number . . . 11
0.4 0-π and π-0 States . . . 12
0.4.1 pedestrian method . . . 12
0.4.2 Pedro's method . . . 17
0.4.3 fermi surface . . . 18
0.4.4 small momentum expansion . . . 19
0.4.5 Continuum Model/ Gradient Expansion . . . 20
0.5 0-π State on Strained Lattice . . . 22
0.5.1 Basics of Strain . . . 22
0.5.2 Straining the 0-π-State . . . 23
0.6 Strain Engineering and other Games with the 0-π-State . . . 24
0.6.1 Uniform Hopping . . . 24
0.6.2 Staggered Hopping . . . 29
0.6.3 On-Site Potential . . . 29

0.1 Construction of Hamiltonian

rename this section to something like model?

0.1.1 Construction of the Hamiltonian

The Hamiltonian is

$$\mathcal{H} = \sum_{\langle ij \rangle} J_{ij} \left(\vec{S}_i \cdot \vec{S}_j + \kappa \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \right) \tag{1}$$

Using the Abrikosov representation of the spin in terms of fermions:

$$\vec{S}_i = \sum_{\alpha\beta} f^\dagger_{i\alpha} \frac{\vec{\sigma}}{2} f_{i\beta} \tag{2}$$

$$\begin{aligned} \Rightarrow \vec{S}_i \cdot \vec{S}_j &= \frac{1}{4} \sum_{\alpha\beta\gamma\delta} f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\gamma} f_{j\delta} \underbrace{\vec{\sigma}_{\alpha\beta} \vec{\sigma}_{\gamma\delta}}_{=\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}} \\ &= \frac{1}{4} \sum_{\alpha\beta} 2 f^\dagger_{i\alpha} f_{i\beta} f^\dagger_{j\beta} f_{j\alpha} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \end{aligned}$$

Move the last annihilation operator in the first term to the second postion: use $f^\dagger_{j\beta} f_{j\alpha} = \delta_{\alpha\beta} - f_{j\alpha} f^\dagger_{j\beta}$ and acquire 1 minus signs. After that bring the other annihilation operator to the last position which gives another minus.

$$\vec{S}_i \cdot \vec{S}_j = \frac{1}{4} \sum_{\alpha\beta} -2 f^\dagger_{i\alpha} f_{j\alpha} f^\dagger_{j\beta} f_{j\beta} + 2 f_{i\alpha} f_{i\beta} \delta_{\alpha\beta} - f^\dagger_{i\alpha} f_{i\alpha} f^\dagger_{j\beta} f_{j\beta} \tag{3}$$

The second term is n_i and the third is $n_i n_j$. For the first term I introduce the shorthand

$$\chi_{ij} = \frac{1}{2} \sum_{\alpha} f^\dagger_{i\alpha} f_{j\alpha} \tag{4}$$

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{2} n_i - \frac{1}{4} n_i n_j \tag{5}$$

Using the constaint $n_i = n_j = 1$, the final expression is

$$\vec{S}_i \cdot \vec{S}_j = -2 \chi_{ij} \chi_{ji} + \frac{1}{4} \tag{6}$$

Relevant for the Hamiltonian is also the term:

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 = \frac{1}{16} - \chi_{ij} \chi_{ji} + 4 \left(\chi_{ij} \chi_{ji} \right)^2 \tag{7}$$

The mean field decoupling works like

$$\chi_{ij} \chi_{ji} \approx \langle \chi_{ij} \rangle \chi_{ji} + \langle \chi_{ji} \rangle \chi_{ij} - |\langle \chi_{ij} \rangle|^2 \tag{8}$$

$$\left(\chi_{ij} \chi_{ji} \right)^2 \approx 2 |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + 2 |\langle \chi_{ij} \rangle|^2 \langle \chi_{ji} \rangle \chi_{ij} - 3 |\langle \chi_{ij} \rangle|^4 \tag{9}$$

$$\Rightarrow \vec{S}_i \cdot \vec{S}_j \approx -2 \langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + 2 |\langle \chi_{ij} \rangle|^2 + \frac{1}{4} \tag{10}$$

$$\Rightarrow \left(\vec{S}_i \cdot \vec{S}_j \right)^2 \approx -\langle \chi_{ij} \rangle \chi_{ji} + 8 |\langle \chi_{ij} \rangle|^2 \langle \chi_{ij} \rangle \chi_{ji} + \text{h.c.} + |\langle \chi_{ij} \rangle|^2 - 12 |\langle \chi_{ij} \rangle|^4 + \frac{1}{16} \tag{11}$$

Maybe comment of the fact that the mean-field parameters know about the fermi sea and thereby about many states. In this sense the states do speak with each other. Bringing everything together:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} J_{ij}\Big[&\left(-1-\kappa/2+4\kappa|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\ &+\left(2+\kappa-12\kappa|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2+\frac{1}{4}+\frac{\kappa}{16}\Big]\end{aligned} \tag{12}$$

It is sometimes handy to define

$$\tilde{J}_{ij} = J_{ij}\left(1+\kappa/2\right), \quad \tilde{\kappa} = \kappa/(1+\kappa/2) \tag{13}$$

I drop the constant:

$$\begin{aligned}\mathcal{H} \approx \sum_{\langle ij\rangle} \tilde{J}_{ij}\Big[&-\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{h.c}\\ &+2\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\Big]\end{aligned} \tag{14}$$

This complies with the result obtained here [2]. Hence, the mean field produces the effective hopping amplitude t^{MF}:

$$t^{\text{MF}}_{ji} = -\tilde{J}_{ij}\langle\chi_{ij}\rangle\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right) = -J_{ij}\langle\chi_{ij}\rangle\left(1+\kappa/2-4\kappa|\langle\chi_{ij}\rangle|^2\right). \tag{15}$$

$$\Rightarrow \mathcal{H} \approx \sum_{\langle ij\rangle} t^{\text{MF}}_{ji}\left(f^\dagger_{j\uparrow}f_{i\uparrow}+f^\dagger_{j\downarrow}f_{i\downarrow}\right)+\text{const.} \tag{16}$$

This is pretty interesting: the hopping-amplitude $i \to j$ is directly proportional to the hopping-amplitude $j \to i$. Here are a few ideas how to make use/discuss this insight:

- Maybe one can also make the argument that we start with a "symmetric" model, i.e. a model wohtout internal sense of direction, a feature that is now preserved in the Mean field Hamiltonian.
- Or maybe, even more suffisticated, one could make argument using strauctures of the spin-singlet states.
- Maybe one can use this fact to siplify the Mean-field equation?

0.1.2 Continuity equation and probability current

The probability (/particle) density on site k is

$$n_k = \sum_\sigma f^\dagger_{k\sigma}f_{k\sigma}. \tag{17}$$

Now I derive the corresponding probability current density:

$$\frac{\partial n_k}{\partial t} = -i\sum_\sigma\left([f^\dagger_{k\sigma},H]f_{k\sigma}+f^\dagger_{k\sigma}[f_{k\sigma},H]\right) \tag{18}$$

$$\begin{aligned}[f^\dagger_{k\sigma},H]f_{k\sigma} &= -\sum_\delta t_{k+\delta;k}f^\dagger_{k+\delta;\sigma}f_{k\sigma}-\mu_k f^\dagger_{k\sigma}f_{k\sigma}\\ f^\dagger_{k\sigma}[f_{k\sigma},H] &= \sum_\delta t_{k;k+\delta}f^\dagger_{k\sigma}f_{k+\delta;\sigma}+\mu f^\dagger_{k\sigma}f_{k\sigma}\end{aligned} \tag{19}$$

$$\Rightarrow \frac{\partial n_k}{\partial t} = -\sum_\delta i \left(\sum_\sigma t_{k,k+\delta} f^\dagger_{k\sigma} f_{k+\delta;\sigma} - t_{k+\delta,k} f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right) \tag{20}$$

Define

$$J_{k\to k+\delta} \equiv i \left(\sum_\sigma t_{k,k+\delta} f^\dagger_{k\sigma} f_{k+\delta;\sigma} - t_{k+\delta,k} f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right) \tag{21}$$

$J_{k\to k+\delta}$ is the (directed!) current density along the bond $\vec{R}_k \to \vec{R}_{k+\delta}$. The discrete (/lattice) continuity equation reads

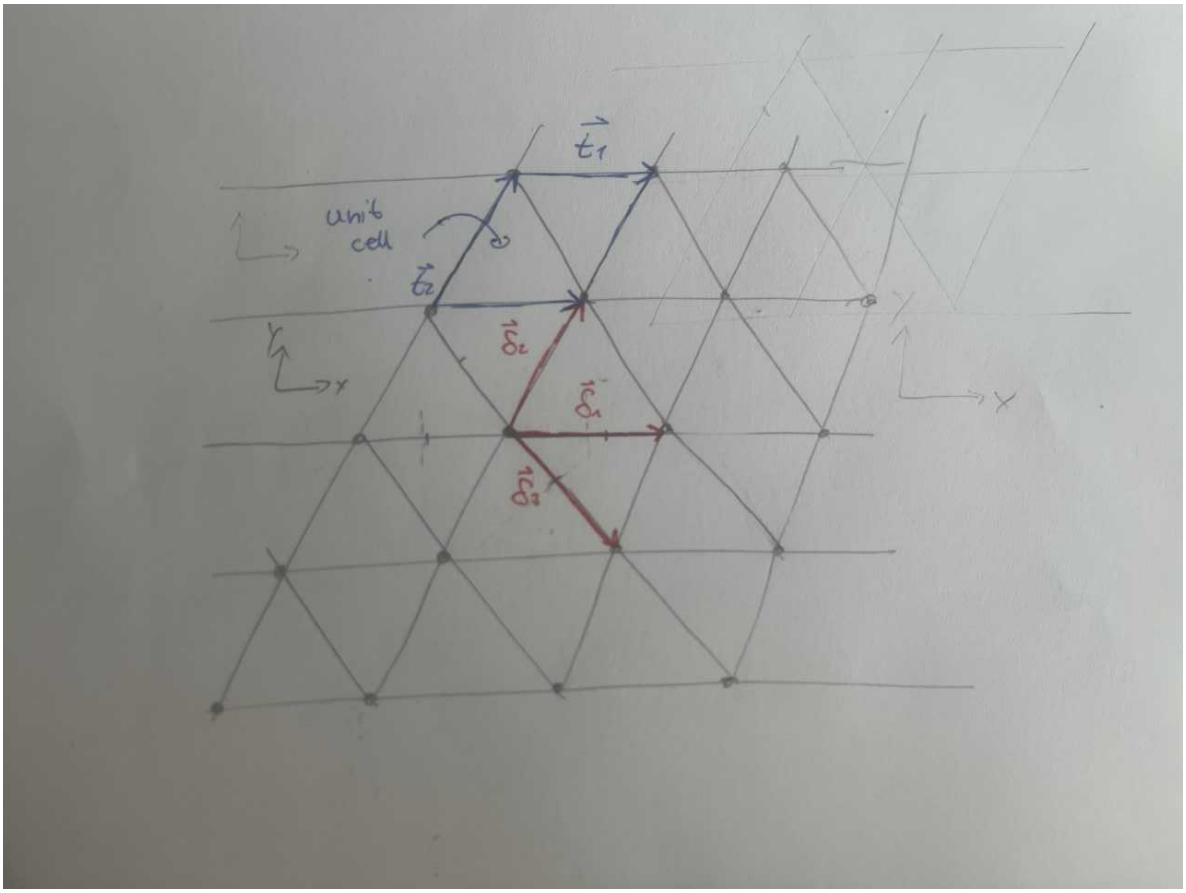
$$\frac{\partial n_k}{\partial t} = -\sum_\delta J_{k\to k+\delta}. \tag{22}$$

Something interesting happens when calculating $\langle J_{k\to k+\delta} \rangle$:

$$\begin{aligned}
\langle J_{k\to k+\delta} \rangle &= i \left(t_{k,k+\delta} \left\langle \sum_\sigma f^\dagger_{k\sigma} f_{k+\delta;\sigma} \right\rangle - t_{k+\delta,k} \left\langle \sum_\sigma f^\dagger_{k+\delta;\sigma} f_{k\sigma} \right\rangle \right) \\
&= 2i \left(t_{k,k+\delta} \langle \chi_{k,k+\delta} \rangle - t_{k+\delta,k} \langle \chi_{k+\delta,k} \rangle \right) \\
&= -2i \tilde{J}_{k;k+\delta} \left(1 - 4\tilde{\kappa} |\langle \chi_{k,k+\delta} \rangle|^2 \right) \left(\langle \chi_{k+\delta;k} \rangle \langle \chi_{k;k+\delta} \rangle - \langle \chi_{k;k+\delta} \rangle \langle \chi_{k+\delta;k} \rangle \right) \\
&= 0.
\end{aligned} \tag{23}$$

Apparently, the current on each bond vanishes in the ground state by construction. This still leaves the possibilitie that single states have non-zero probability currents. However, it remains to investigate how this is possible for e.g. states with a singular (topologically protected) edge mode. One would guess that this feature would lead to a non-zero current in the ground state.

0.2 0-0 and π-π States



This system is a simple triangular Bravais lattice with primitve translations

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1/2 \\ \sqrt{3}/2 \end{pmatrix}. \tag{24}$$

0.2.1 Pedestrian method

The tight-binding Hamiltonian is:

$$\begin{aligned} H &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_i \left(f^\dagger_{i\uparrow} f_{i\uparrow} + f^\dagger_{i\downarrow} f_{i\downarrow} - 1 \right) + \text{const.} \\ &= \sum_{i,\delta,\sigma} \frac{t}{2} \left(f^\dagger_{i\sigma} f_{i+\delta\,\sigma} + \text{h.c.} \right) + \mu \sum_{i,\sigma} f^\dagger_{i\sigma} f_{i\sigma} - \mu N + \text{const.}. \end{aligned} \tag{25}$$

i runs over sites sites and δ over nearest neighbours of a generic site. I include the absolute shift for the sake of consistency. The factor $1/2$ is also included for consistency. The constant is a product of mean field theory and reads

$$\frac{1}{2} \sum_{i,\delta} 2\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} |^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 = 6N \left(\tilde{J} \left(1 - 6\tilde{\kappa} |\langle \chi_{i,i+\delta} \rangle|^2 \rangle \right) |\langle \chi_{i,i+\delta} \rangle|^2 \right) \tag{26}$$

The second step can be understood in at least two ways; a) each site has 6 neighbours, or b) there are bond per primitive unit cell 1 of which is completely inside the unit cell and contributes a 2 and the

other 4 shared between 2 unit cells thereby contributing a 1. All bonds are equivalent of course. The afformentioned six nearest neighbours lie at the (relative) positions

$$\vec{\delta_1} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \; \vec{\delta_2} = \begin{pmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{pmatrix} \; \vec{\delta_3} = \begin{pmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{pmatrix}$$

and $-\vec{\delta_1}$, $-\vec{\delta_2}$, $-\vec{\delta_3}$. I will start with the $0-0$ state. To advance, do a Fourier transform:

$$f_{i\sigma} = \frac{1}{\sqrt{N}} \sum_k e^{i\vec{k}\vec{R}_i} c_{k\sigma}. \tag{27}$$

$$\begin{aligned} \Rightarrow H &= \sum_{q,\sigma} \frac{t}{2} \left(c^\dagger_{q\sigma} c_{q\sigma} \left(\sum_\delta e^{i\vec{q}\vec{\delta}} + \mu \right) + \text{h.c.} \right) - \mu N \\ &= \sum_{q,\sigma} \left(t \sum_\delta \cos\left(\vec{q}\vec{\delta}\right) + \mu \right) c^\dagger_{q\sigma} c_{q\sigma} - \mu N \end{aligned} \tag{28}$$

Using first the symmetrie of the cosine and then the sum identity for the cosine the dispersion reads

$$\begin{aligned} \varepsilon(\vec{q}) &= 2t \left[\cos(q_x) + \cos\left(\frac{1}{2}(q_x + \sqrt{3} q_y) \right) + \cos\left(\frac{1}{2}(q_x - \sqrt{3} q_y) \right) \right] + \mu \\ &= 2t \left[\cos(q_x) + 2\cos\left(\frac{q_x}{2} \right) \cos\left(\frac{\sqrt{3} q_y}{2} \right) \right] + \mu \end{aligned} \tag{29}$$

features:

- $\varepsilon(\vec{q}) \in [-3t, 6t] + \mu \Rightarrow \mu \in [-6t, 3t]$
- symmetries $q_x \to -q_x$ and $q_y \to -q_y$ (+ should have some rotational symmetrie and (thereby associated) inflections along other axes) could check this
- periodicity: 4π in q_x, $(4\pi)/\sqrt{3}$ in q_y

The mean field parameters are

$$\langle \chi_{ij} \rangle = \frac{1}{2} \langle f^\dagger_{i\uparrow} f_{j\uparrow} + f^\dagger_{i\downarrow} f_{j\downarrow} \rangle = \langle f^\dagger_{i\uparrow} f_{j\uparrow} \rangle \tag{30}$$

Simplify the expression by Fourier transform:

$$\begin{aligned} \langle \chi_{ij} \rangle &= \frac{1}{N} \sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{k}\vec{R}_i)} \langle c^\dagger_p c_q \rangle \\ &= \frac{1}{N} \sum e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \end{aligned} \tag{31}$$

Take the thermodynamic limit $\sum_k \to (N\Omega)/(2\pi)^2 \int dk$ (Ω is the unit cell volume).

$$\Rightarrow \langle \chi_{ij} \rangle = \Omega \int_{1.\text{BZ}} \frac{d^2 q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} f(\varepsilon(\vec{q})) \tag{32}$$

It looks like the model allows for different χ_{ij} in each direction $\vec{R}_j - \vec{R}_i$ (there are 3 of those)?

From this expression, recover the on-site particle number by setting $i = j$ and accounting for the factor $1/2$ in the definition of χ_{ij}:

$$\langle N_i \rangle = 2\Omega \int \frac{d^2q}{(2\pi)^2} f(\varepsilon(\vec{q})). \tag{33}$$

By construction the particle number is uniform.
Some elaborations on the integral domain follow. The primite translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi \begin{pmatrix} 0 \\ 2/\sqrt{3} \end{pmatrix}, \quad \vec{b}_2 = 2\pi \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \tag{34}$$

I do a basis transformation onto thos axes defined by

$$\begin{pmatrix} k_x \\ k_y \end{pmatrix} = \begin{pmatrix} 1/2 & \sqrt{3}/2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_x \\ q_y \end{pmatrix} \tag{35}$$

$$\Leftrightarrow \begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 2/\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{36}$$

$$\Rightarrow \det J = -2/\sqrt{3} \tag{37}$$

$$\Rightarrow \varepsilon(\vec{k}) = 2t\left(\cos(k_x) + \cos(k_y) + \cos(k_x - k_y)\right) + \mu \tag{38}$$

The first Brillouin zone is $\{k_x, k_y\} \in [-\pi, \pi]^2$. The unit cell volume is $\Omega = \sqrt{3}/2$. Consistency check: $|\det(\mathrm{J})| = \Omega^{-1}$, since the substitution simply transforms the unti cell to a square of unit volume. So the integrals that have to be evaluated are of the form

$$\langle N_i \rangle = 2 \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} f(\varepsilon(\vec{k}), \mu), \tag{39}$$

$$\langle \chi_{ij} \rangle = \int_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\Big[i\left(k_y, (1/\sqrt{3})(2k_x - k_y)\right) \cdot (\vec{R}_j - \vec{R}_i)\Big] f(\varepsilon(\vec{k}), \mu) \tag{40}$$

Check that under a particle hole transformation chi changes sign, sign of t is doing just that and this brings the *pi*-flux phase.
The condition is $\langle N_i \rangle \stackrel{!}{=} 1 \; \forall i$. By setting setting $t = -1$, the chemical potential is numerically determined in units of t using mathematica. I choose $t = -1$ because I anticipate t to be negative for a positive χ. (This may not be true depending on κ and χ, but as long as it is self-consistent evrything is fine.) For the numerical integration I used a Fermi-function with a small temperature ($\beta = 200t$) instead of a step-function, since the ladder is not analytical and causes problems for the numerical algorithm. The result is $\mu = -0.834737\ t$. The bond-parameters are calulcated the same way, the result is $\langle \chi_{ij} \rangle = 0.164713$.

Now, it is possible to express μ only in units of J. Use the formula

$$t_{ij} = -J\langle \chi_{ij} \rangle^* (1 + \kappa/2 - 4\kappa\, |\langle \chi_{ij} \rangle|^2). \tag{41}$$

Set $\kappa = 10$:

$$t_{ij} = -0.8095\ J \tag{42}$$

$$\mu = -0.67574\ J \tag{43}$$

$$\langle \chi_{ij} \rangle = 0.164713 \tag{44}$$

The energy density (that is also the free energy desnity at $T = 0$) is

$$\frac{\langle H\rangle}{N} = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon(\vec{k})f(\varepsilon(\vec{k})) - \mu + 6\tilde{J}\left(1 - 6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \; = -0.888304\,J \tag{45}$$

Observtaion; the chemical potential drops out of the Free energy.

One can also directly calculate results for the state with all fluxes π by performing a particle-hole-transformation:

$$t \to -t$$
$$\mu \to -\mu \tag{46}$$
$$\Rightarrow \varepsilon(\vec{k}) \to -\varepsilon(\vec{k}) \tag{47}$$
$$\Rightarrow f(\varepsilon(\vec{k})) \to 1 - f(\varepsilon(\vec{k}) \tag{48}$$

It can be shown that

$$\chi \to -\chi. \tag{49}$$

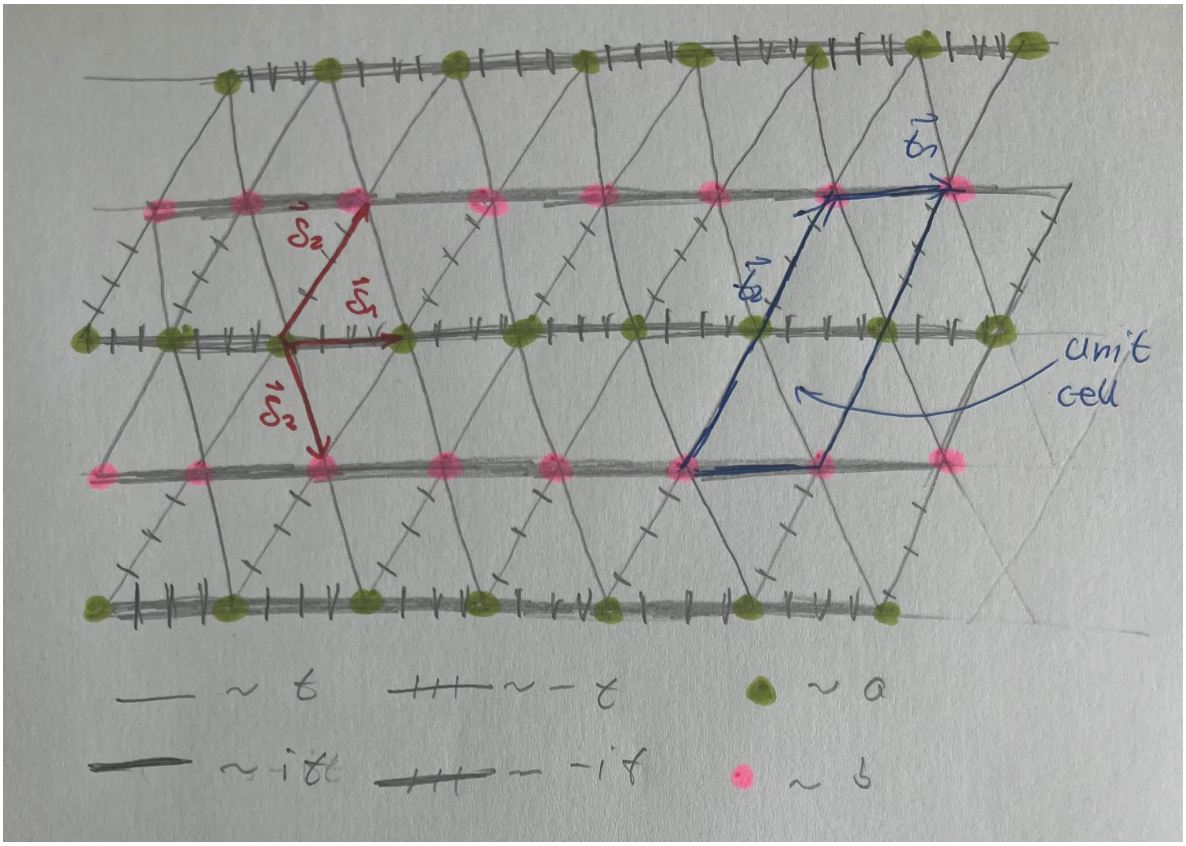
The result is:

$$\frac{\langle H\rangle_\pi}{N} = -0.888302\,J \tag{50}$$

0.2.2 Pedro's method

The longer explenation for this method is in the next section. Here I will only say that the free energy is the same except for γ, which gets replaced by a factor ϕ (as mentioned above the chemical potentioal does not contribute to the free energy):

0.3 $\pi/2$-$\pi/2$ and $-\pi/2$ - $-\pi/2$ States



0.3.1 Pedestrian Method

This is very similar to the $\pi - 0-$flux-state. The bare Hamiltonian is

$$\begin{aligned} H = \sum_{i,\sigma} \Bigg(\frac{t}{2} \Big[& - i a^\dagger_{i\sigma} a_{i+\delta_1;\sigma} + i a^\dagger_{i\sigma} a_{i-\delta_1;\sigma} - a^\dagger_{i\sigma} b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \\ & + i b^\dagger_{i\sigma} b_{i+\delta_1;\sigma} - i b^\dagger_{i\sigma} b_{i-\delta_1;\sigma} + b^\dagger_{i\sigma} a_{i+\delta_2;\sigma} - b^\dagger_{i\sigma} a_{i-\delta_2;\sigma} + a^\dagger_{i\sigma} b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma} b_{i-\delta_3;\sigma} \Big] \\ & + \mu_a a^\dagger_{i\sigma} a_{i\sigma} + \mu_b b^\dagger_{i\sigma} b_{i\sigma} \Bigg) - \frac{N(\mu_a + \mu_b)}{2} \end{aligned} \tag{51}$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices.

$$\Rightarrow H = \sum_{q,\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} \end{pmatrix} \begin{pmatrix} 2t \sin\left(\vec{q}\vec{\delta}_1\right) + \mu_a & 2t \cos\left(\vec{q}\vec{\delta}_3\right) - 2it \sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t \cos\left(\vec{q}\vec{\delta}_3\right) + 2it \sin\left(\vec{q}\vec{\delta}_2\right) & -2t \sin\left(\vec{q}\vec{\delta}_1\right) + \mu_b \end{pmatrix} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix}. \tag{52}$$

Maybe I should show explicitly that $\mu = 0$, I guess $\mu_a = -\mu_b = \mu$ is clear
Using $\mu_a = \mu_b = 0$:

$$\vec{h} = 2t \begin{pmatrix} \cos\left(\vec{q}\vec{\delta}_1\right) \\ \sin\left(\vec{q}\vec{\delta}_2\right) \\ \sin\left(\vec{q}\vec{\delta}_3\right) \end{pmatrix} \mapsto 2t \begin{pmatrix} \cos(k_y - k_x/2) \\ \sin(k_x/2) \\ \sin(k_y) \end{pmatrix} \tag{53}$$

$$\varepsilon(\vec{k})_{\pm} = \pm 2|t|\sqrt{\sin^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \tag{54}$$

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\sin(k_y))}} \begin{pmatrix} \varepsilon_+ + 2t\sin(k_y) & -2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) \\ 2t\cos(k_y - k_x/2) + 2ti\sin(k_x/2) & \varepsilon_+ + 2t\sin(k_y) \end{pmatrix} \tag{55}$$

$$u_{11}u_{21} = (t/\varepsilon_+)(\cos(k_y - k_x/2) + i\sin(k_x/2)) \tag{56}$$
$$u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t\sin(k_y)) \tag{57}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)(1-f(\varepsilon_+)) + u_{11}^2\right] \tag{58}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y}\left[(1-2u_{11}^2)f(\varepsilon_+) + u_{11}^2\right] \tag{59}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+) - 1) \tag{60}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma}\rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+) - 1) \tag{61}$$

One has to be a bit careful of where the sign of t enters when choosing $t = 1$ for the self consistency. Due to the complex conjugation the σ_z-terms are insensitive to the sign of t, but the (real) σ_x- and σ_y-terms are not. The results (choosing $t = 1$ are):

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma}\rangle = -0.200169i \tag{62}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i-\delta_1\sigma}\rangle = 0.200169i \tag{63}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma}\rangle = 0.200169i \tag{64}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i-\delta_1\sigma}\rangle = -0.200169i \tag{65}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma}\rangle = 0.200169 \tag{66}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i-\delta_3\sigma}\rangle = 0.200169 \tag{67}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma}\rangle = 0.200169 \tag{68}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i-\delta_2\sigma}\rangle = -0.200169 \tag{69}$$

This complies with the cartoon on top. The value for t is

$$|t_{ij}| = \left|J\langle\chi_{ij}\rangle^*(1 + \kappa/2 - 4\kappa|\langle\chi_{ij}\rangle|^2\right| \overset{\kappa=10}{=} 0.88020\, J. \tag{70}$$

The free energy is

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) = -1.24978\ J \quad (71)$$

0.3.2 Pedro's Method

The expression for the free energy is identical to the $0-\pi$-state except that γ gets replaced by a different constant α:

$$\alpha = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\sin^2(k_y)+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \approx 1.20102 \quad (72)$$

Crucally α is larger than gamma. The results for this method are

$$|\chi| = \alpha/6 = 0.20017, \quad (73)$$

$$f_0 = \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\alpha}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right) = -1.24979\,J \quad (74)$$

The methods agree.

0.3.3 Chern number

Here I will calculate the Chern number associated with the state. The relevant formula for a 2-band system is is

$$C = \frac{1}{4\pi}\int\limits_{1.\mathrm{BZ}} d^2q\,\hat{h}\cdot\left(\partial_{q_x}\hat{h}\times\partial_{q_y}\hat{h}\right), \quad (75)$$

where $\hat{h}$ is the normalized Bloch vector (is this the correct terminology?).

$$\vec{h} = 2t\begin{pmatrix}\cos\left(\frac{1}{2}(q_x-\sqrt{3}q_y)\right)\\ \sin\left(\frac{1}{2}(q_x+\sqrt{3}q_y)\right)\\ \sin(q_x)\end{pmatrix} \equiv 2t\begin{pmatrix}\cos(\delta_3)\\ \sin(\delta_2)\\ \sin(\delta_1)\end{pmatrix}, \quad \left|\vec{h}\right| \equiv \varepsilon(\vec{q}) \quad (76)$$

I first simplify 75 a bit using

$$\partial_\alpha\hat{h} = \frac{1}{\left|\vec{h}\right|}\left(\partial_\alpha\vec{h} - \left(\hat{h}\cdot\partial_\alpha\vec{h}\right)\hat{h}\right). \quad (77)$$

When applying this to 75, 3 out of the for terms that appear are proportional to $\hat{h}\times\hat{h} = \vec{0}$.

$$\Rightarrow C = \frac{1}{4\pi}\int\limits_{1.BZ} d^2q\,[\varepsilon(\vec{q})]^{-3}\vec{h}\cdot\left(\partial_{q_x}\vec{h}\times\partial_{q_y}\vec{h}\right) \quad (78)$$

$$\partial_{q_x}\vec{h} = t\begin{pmatrix}-\sin(\delta_3)\\ \cos(\delta_2)\\ 2\cos(\delta_1)\end{pmatrix}, \qquad \partial_{q_y}\vec{h} = \sqrt{3}t\begin{pmatrix}\sin(\delta_1)\\ \cos(\delta_2)\\ 0\end{pmatrix} \quad (79)$$

$$\begin{aligned}\Rightarrow \vec{h}\cdot\left(\partial_{q_x}\vec{h}\times\partial_{q_y}\vec{h}\right) &= 4\sqrt{3}t^3\left(-\cos(\delta_1)\cos(\delta_2)\cos(\delta_3)+\cos(\delta_1)\sin(\delta_2)\sin(\delta_3)-\sin(\delta_1)\cos(\delta_2)\sin(\delta_3)\right)\\ &= 4\sqrt{3}t^3\left(\sin(\delta_1)\sin(\delta_2)\cos(\delta_3)-1\right) \quad (80)\end{aligned}$$

$$\Rightarrow C = \frac{\sqrt{3}}{8\pi} \int\limits_{1.\mathrm{BZ}} d^2q \frac{\sin(\delta_1)\sin(\delta_2)\cos(\delta_3) - 1}{\left(\sin^2(\delta_1) + \sin^2(\delta_2) + \cos^2(\delta_3)\right)^{3/2}} \tag{81}$$

For evaluation I use the transformation:

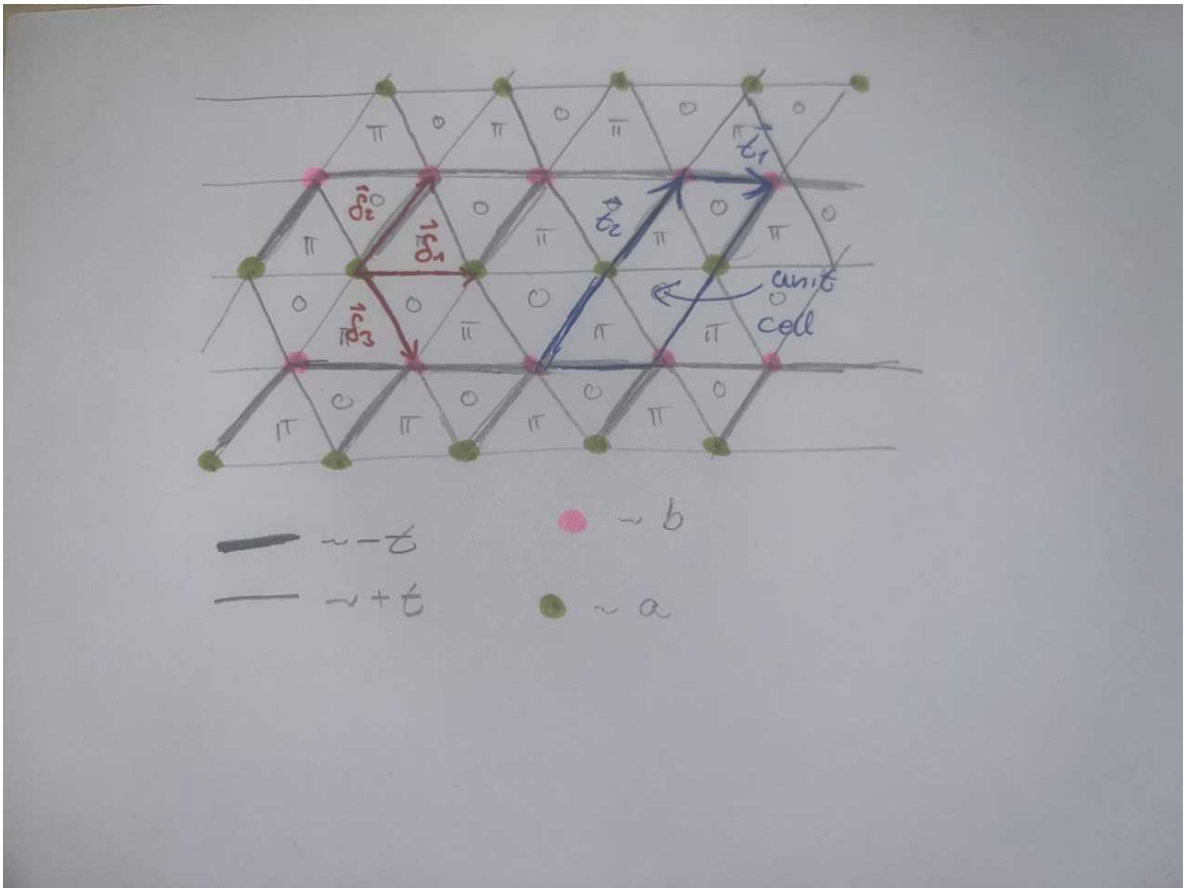
$$\begin{pmatrix} q_x \\ q_y \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1\sqrt{3} & -1/\sqrt{3} \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix} \tag{82}$$

$$\Rightarrow C = \frac{1}{8\pi} \int\limits_{[-\pi,\pi]^2} d^2k \frac{\sin(k_y)\sin(k_y - k_x/2)\cos(k_x/2) - 1}{\left(\sin^2(k_y) + \sin^2(k_y - k_x/2) + \cos^2(k_x/2)\right)^{3/2}} \tag{83}$$

The result is $C = -1$.

0.4 0-π and π-0 States

0.4.1 pedestrian method



The system is a triangular bravais lattice with a two-atomic basis. The primitve translations are

$$\vec{a}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{a}_2 = \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \tag{84}$$

Hence, two types of lattice sites exist, a and b (see sketch). Also χ_{ij}, and thereby t_{ij} can change sign (see sketch). The Hamiltonian is

$$H = \frac{1}{2}\sum_{\langle i,j\rangle,\sigma}\left(ta^\dagger_{i\sigma}a_{j\sigma} - tb^\dagger_{i\sigma}b_{j\sigma} + t^{(ab)}_{ij}a^\dagger_{i\sigma}b_{j\sigma} + \text{h.c.}\right) + \mu_a\sum_{i,\sigma}\left(a^\dagger_{i\sigma}a_{i\sigma} - 1\right) + \mu_b\sum_{i,\sigma}\left(b^\dagger_{i\sigma}b_{i\sigma} - 1\right) + \text{const.} \tag{85}$$

$$\begin{aligned} H = \frac{t}{2}\sum_{i\sigma}\Big(& a^\dagger_{i\sigma}a_{i+\delta_1;\sigma} + a^\dagger_{i\sigma}a_{i-\delta_1;\sigma} - a^\dagger_{i\sigma}b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma}b_{i-\delta_2;\sigma} + a^\dagger_{i\sigma}b_{i+\delta_3;\sigma} + a^\dagger_{i\sigma}b_{i-\delta_3;\sigma} \\ & - b^\dagger_{i\sigma}b_{i+\delta_1;\sigma} - b^\dagger_{i\sigma}b_{i-\delta_1;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_2;\sigma} - b^\dagger_{i\sigma}a_{i-\delta_2;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_3;\sigma} + b^\dagger_{i\sigma}a^\dagger_{i-\delta_3;\sigma} \\ & + \mu_a a^\dagger_{i\sigma}a_{i\sigma} + \mu_b b^\dagger_{i\sigma}b_{i\sigma}\Big) - \frac{N(\mu_a+\mu_b)}{2} \end{aligned} \tag{86}$$

The site index notation is not ideal, as a and b sites by definition cant have the same site indices. The constant is the same as for the 0-flux case, provided that the absolute value of $\chi_{i,i+\delta}$ is uniform.

$$\frac{1}{2}\sum_{i,\delta} 2\tilde{J}\left(1 - 6|\langle\chi_{i,i+\delta}|^2\rangle\right)|\langle\chi_{i,i+\delta}\rangle|^2 = 6N\tilde{J}\left(1 - 6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2 \tag{87}$$

This is not quite wright yet. The Fourier transformation has to be modified bc there are only N/2 of a and b-sites respectively

$$a_{i\sigma} = \sqrt{\frac{2}{N}}\sum_k e^{i\vec{k}\vec{R}_i}a_{k\sigma}, \tag{88}$$

and the same for $b_{i\sigma}$. The Hamiltonian (without the constants) becomes

$$H = \sum_{q,\sigma}\begin{pmatrix}a^\dagger_{q\sigma} & b^\dagger_{q\sigma}\end{pmatrix}\begin{pmatrix}2t\cos\left(\vec{q}\vec{\delta}_1\right) + \mu_a & 2t\cos\left(\vec{q}\vec{\delta_3}\right) - 2it\sin\left(\vec{q}\vec{\delta}_2\right) \\ 2t\cos\left(\vec{q}\vec{\delta}_3\right) + 2it\sin\left(\vec{q}\vec{\delta}_2\right) & -2t\cos\left(\vec{q}\vec{\delta}_1\right) + \mu_b\end{pmatrix}\begin{pmatrix}a_{q\sigma}\\ b_{q\sigma}\end{pmatrix}. \tag{89}$$

The primitive translations of the reciprocal lattice are

$$\vec{b}_1 = 2\pi\begin{pmatrix}0\\ 1/\sqrt{3}\end{pmatrix}, \quad \vec{b}_2 = 2\pi\begin{pmatrix}1\\ -1/\sqrt{3}\end{pmatrix} \tag{90}$$

.
The suitable transformation is

$$\begin{pmatrix}k_x\\ k_y\end{pmatrix} = \begin{pmatrix}1 & \sqrt{3}\\ m1 & 0\end{pmatrix}\begin{pmatrix}q_x\\ q_y\end{pmatrix} \tag{91}$$

$$\Leftrightarrow \begin{pmatrix}q_x\\ q_y\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1/\sqrt{3} & -1/\sqrt{3}\end{pmatrix}\begin{pmatrix}k_x\\ k_y\end{pmatrix} \tag{92}$$

$$\Rightarrow \det(J) = 1/\sqrt{3} \tag{93}$$

This yields

$$\cos\left(\vec{q}\vec{\delta}_1\right) \to \cos(k_y)$$

$$\sin\left(\vec{q}\vec{\delta}_2\right) \to \sin(k_x/2) \tag{94}$$

$$\cos\left(\vec{q}\vec{\delta}_3\right) \to \cos(k_y - k_x/2) \tag{95}$$

$$\varepsilon(\vec{k})_\pm = \mu_a+\mu_b\pm\sqrt{(\mu_a-\mu_b)^2+2t\cos(k_y)(\mu_a-\mu_b)+4t^2\cos^2(k_y)+4t^2\cos^2(k_y-k_x/2)+4t^2\sin^2(k_x/2)} \tag{96}$$

Apparently, half filling is achieved globally, iff $\mu_a = -\mu_b \equiv \mu$. (The spectrum is symmetric around $\varepsilon = 0$ in this case.) In order to fulfill the local constraint one still has to calculate the local expectation values. It would be weird however, if the chemical potential differed between a and b sites (-¿ charge density waves?).

$$\Rightarrow \varepsilon(\vec{k})_\pm = \pm 2|t|\sqrt{(\mu/(2t)+\cos(k_y))^2+\cos^2(k_y-k_x/2)+\sin^2(k_x/2)} \tag{97}$$

It is correct that ε only depends on the absolute value of t, it looks like this would completely neutrilise the particle-hole transformation.The periodicity is a bit weird here, since the expression is not periondic in 2π in every direction as one would expect. I am prettys rue this comes from the fact that the distance between NN is the same as for the 0-flux case but the Brilouin zone got cut in hoalf in one direction. represent hamiltonian as

$$H = \sum_{q,\sigma} \begin{pmatrix} a^\dagger_{q\sigma} & b^\dagger_{q\sigma} \end{pmatrix} \vec{h}\cdot\vec{\sigma} \begin{pmatrix} a_{q\sigma} \\ b_{q\sigma} \end{pmatrix} \tag{98}$$

$$\vec{h} = \begin{pmatrix} 2t\cos(k_y-k_x/2)) \\ 2t\sin(k_x/2) \\ 2t\cos(k_y)+\mu \end{pmatrix} \tag{99}$$

Then $\varepsilon(\vec{q})_\pm = \pm|\vec{h}|$ and

$$\frac{1}{\sqrt{2\varepsilon_+(\varepsilon_+ + 2t\cos(k_y)+\mu)}} \begin{pmatrix} \varepsilon_+ + 2t\cos(k_y)+\mu & -2t\cos(k_y-k_x/2)+2ti\sin(k_x/2) \\ 2t\cos(k_y-k_x/2)+2ti\sin(k_x/2) & \varepsilon_+ + 2t\cos(k_y)+\mu \end{pmatrix} \tag{100}$$

is the transformation matrix U. It has the properties that it is unitary and has determinant one. The expectation values are more involved bc of said matrix transformation. For example

$$\begin{aligned} \langle a^\dagger_{i\uparrow} b_{j\uparrow}\rangle &\overset{\text{FT}}{=} \frac{2}{N}\sum_{q,p} e^{i(\vec{q}\vec{R}_j - \vec{p}\vec{R}_i)} \underbrace{\langle a^\dagger_p b_q\rangle}_{\propto\delta_{q,p}} \\ &= \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\langle a^\dagger_q b_q\rangle \\ &\overset{\text{BT}}{=} \frac{2}{N}\sum_q e^{i\vec{q}(\vec{R}_j-\vec{R}_i)}\left(u^*_{11}u_{21}f(\varepsilon_+)+u^*_{12}u_{22}f(\varepsilon_-)\right) \end{aligned} \tag{101}$$

Simplify this using

1. $\varepsilon_+ = -\varepsilon_- \Rightarrow f(\varepsilon_+)+f(\varepsilon_-) = 1$
2. $u_{11} \in \mathbb{R}$
3. $u_{11} = u_{22}$
4. $u^*_{12} = -u_{21}$
5. $|u_{11}|^2 + |u_{12}|^2 = 1$ (normalization of eigenvectors)

$$
\begin{aligned}
u_{11}^* u_{21} f(\varepsilon_+) + u_{12}^* u_{22} f(\varepsilon_-) &\overset{1.}{=} (u_{11}^* u_{21} - u_{22} u_{12}^*) f(\varepsilon_+) + u_{22} u_{12}^* \\
&\overset{2.,3.,4.}{=} 2u_{11} u_{21} f(\varepsilon_+) - u_{11} u_{21} \\
&= u_{11} u_{21} (2f(\varepsilon_+) - 1)
\end{aligned} \tag{102}
$$

Due to the unit cell doubling the thermodynamic limit has to be slightly modified $\sum_k \to ((\Omega N)/2)/(2\pi)^2 \int dk$. (Each unit cell has 2 sites, this is why the factor 1/2 appears.)

$$
\begin{aligned}
\Rightarrow \langle a_{i\uparrow}^\dagger b_{j\uparrow} \rangle &= \Omega \int\limits_{1.\mathrm{BZ}} \frac{d^2q}{(2\pi)^2} e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} u_{11} u_{21} (2f(\varepsilon_+) - 1) \\
&= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21} (2f(\varepsilon_+) - 1)
\end{aligned} \tag{103}
$$

$$
u_{11} u_{21} = (t/\varepsilon)(\cos(k_y - k_x/2) + i \sin(k_x/2)) \tag{104}
$$

Now for the other expectation values:

$$
\Rightarrow \langle b_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] u_{11} u_{21}^* \, (2f(\varepsilon_+) - 1) \tag{105}
$$

$$
\begin{aligned}
\langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle = ... &= \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[|u_{11}|^2 f(\varepsilon_+) + |u_{12}|^2 f(\varepsilon_-) \right] \\
&\overset{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[|u_{12}|^2 - |u_{11}|^2) f(\varepsilon_-) + |u_{11}|^2 \right] \\
&\overset{5.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[(1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2 \right] \\
&\overset{1.}{=} \frac{2}{N} \sum_q e^{i\vec{q}(\vec{R}_j - \vec{R}_i)} \left[(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2 \right]
\end{aligned} \tag{106}
$$

$$
\Rightarrow \langle a_{i\uparrow}^\dagger a_{j\uparrow} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] \left[(1 - 2u_{11}^2)(1 - f(\varepsilon_+)) + u_{11}^2 \right] \tag{107}
$$

$$
\Rightarrow \langle b_{i\uparrow}^\dagger b_{j\uparrow} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} \exp\left[i \left(k_y, 1/\sqrt{3}(k_x - k_y) \right)^T (\vec{R}_j - \vec{R}_i) \right] \left[(1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2 \right] \tag{108}
$$

$$
u_{11}^2 = (1/2\varepsilon_+)(\varepsilon_+ + 2t \cos(k_y) + \mu) \tag{109}
$$

$$
\langle N^a \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_-) + u_{11}^2 \tag{110}
$$

$$
\langle N^b \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} (1 - 2u_{11}^2) f(\varepsilon_+) + u_{11}^2 \tag{111}
$$

The constraint $\langle N_i^a \rangle = 1 = \langle N_i^b \rangle$ should, as gloal halffiling is already ensured and $\langle N^a \rangle$ and $\langle N^b \rangle$ are uniform by construction, be equivalent to $\langle N^a \rangle = \langle N^b \rangle$. Using the expressions above:

$$\langle N^a \rangle - \langle N^b \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}(1-2u_{11}^2)(1-2f(\varepsilon_+)) \stackrel{!}{=} 0 \tag{112}$$

Numerics say that $\langle N^a \rangle = \langle N^b \rangle$ for $\mu = 0$ (and therefore $\mu_a = \mu_b = 0$). I feel like there is a more suphisticated argument to be made here, mabye even without the use of mathematica. Maybe one can find some symmetry argument using the form of $1-2f(\varepsilon_+)$.

Sanity check:

$$\langle N^a + N^b \rangle = 2 \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} 1 = 2 \tag{113}$$

Calculating the mean field parameters is less straight forward, as the formulae also depend on the types of sites they connect. The formulae are

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} \left[(1-2u_{11}^2)(1-f(\varepsilon_+)) + u_{11}^2\right] \tag{114}$$

$$\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} \left[(1-2u_{11}^2) f(\varepsilon_+) + u_{11}^2\right] \tag{115}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_3\sigma} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{i(k_y - k_x/2)} u_{11}u_{21}(2f(\varepsilon_+)-1) \tag{116}$$

$$\frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger b_{i+\delta_2\sigma} \rangle = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_x/2} u_{11}u_{21}^*(2f(\varepsilon_+)-1) \tag{117}$$

Consistency check: wehn reversing the direction of the bonds the $\vec{\delta}_2$ bond paramter should obtain a minus sign but the $\vec{\delta}_3$ bond parameter not. I wasnt able to figure this out by just looking at the formulae, so Ill just have to check by looking at the numerics for now.

Another observation:

$$\begin{aligned}
\frac{1}{2}\sum_\sigma \left(\langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma} \rangle + \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma} \rangle\right) &= \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2} e^{ik_y} \\
&= \frac{1}{2\pi} \int\limits_{[-\pi,\pi]} dk_y e^{ik_y} \\
&= \frac{\sin(\pi)}{\pi} \\
&= 0
\end{aligned} \tag{118}$$

$$\Rightarrow \frac{1}{2}\sum_\sigma \langle a_{i\sigma}^\dagger a_{i+\delta_1\sigma} \rangle = -\frac{1}{2}\sum_\sigma \langle b_{i\sigma}^\dagger b_{i+\delta_1\sigma} \rangle, \tag{119}$$

as expected. The numerical results are

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}a_{i+\delta_1\sigma}\rangle = 0.19754 \tag{120}$$

$$\frac{1}{2}\sum_{\sigma}\langle b^{\dagger}_{i\sigma}b_{i+\delta_1\sigma}\rangle = -0.19754 \tag{121}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_3\sigma}\rangle = 0.19754 \tag{122}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i+\delta_2\sigma}\rangle = 0.19754 \tag{123}$$

$$\frac{1}{2}\sum_{\sigma}\langle a^{\dagger}_{i\sigma}b_{i-\delta_2\sigma}\rangle = -0.19754 \tag{124}$$

The corresponding hopping amplitudes are

$$t(\pm 0.19754) = \mp 0.87690J \tag{125}$$

$$\tag{126}$$

At last I will derive the energy density (since I had trouble with this before).

$$\begin{aligned}\frac{\langle H\rangle}{N} &= \frac{1}{N}\left\langle \sum_{q\sigma}\varepsilon_{+\sigma}\alpha^{\dagger}_{+\sigma}\alpha_{+\sigma} + \varepsilon_{-\sigma}\alpha^{\dagger}_{-\sigma}\alpha_{-\sigma}\right\rangle \\ &\overset{\text{th.lim.}}{\longrightarrow} 2\frac{1}{N}\frac{\Omega N}{2}\int\limits_{1.\text{BZ}}\frac{d^2q}{(2\pi)^2}\varepsilon_+ f(\varepsilon_+) + \varepsilon_- f(\varepsilon_-) \\ &= \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}\varepsilon_+(2f(\varepsilon_+)-1)\end{aligned} \tag{127}$$

(the blue 2 in the second line is the spin degeneracy)
Observation: the energy density is invariant under particle-hole transformation

$$\varepsilon_+(2f(\varepsilon_+)-1) \overset{\text{ph-trafo}}{\longrightarrow} \varepsilon_-(2f(\varepsilon_-)-1) = -\varepsilon_+(1-2f(\varepsilon_+)) = \varepsilon_+(2f(\varepsilon_+)-1) \tag{128}$$

The full energy density is:

$$\frac{\langle H\rangle}{N} = \int\limits_{[-\pi,\pi]^2}\frac{d^2k}{(2\pi)^2}\varepsilon_+(\vec{k})(2f(\varepsilon_+(\vec{k}))-1) + 6\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{i,i+\delta}\rangle|^2\right)|\langle\chi_{i,i+\delta}\rangle|^2\right) = -1.2206\ J \tag{129}$$

0.4.2 Pedro's method

Here I apply Pedros method using the terms from my Hamiltonian. The free energy is

$$f_0 \equiv \frac{F_0(\kappa,\chi,T=0)}{N} = \frac{E_0}{N} + \frac{1}{N}\sum_{\vec{k}}\varepsilon(\vec{k}) \tag{130}$$

$$E_0 = 6N\left(\tilde{J}\left(1-6\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)|\langle\chi_{ij}\rangle|^2\right) \tag{131}$$

$$\varepsilon(\vec{k}) = -2\left|-\tilde{J}\left(1-4\tilde{\kappa}|\langle\chi_{ij}\rangle|^2\right)\langle\chi_{ij}\rangle\right|\sqrt{\cos^2(k_1)+\cos^2(k_3)+\sin^2(k_2)} \tag{132}$$

From now on I use the shorthand $|\langle\chi_{ij}\rangle| \equiv \chi$.

$$\begin{aligned} \Rightarrow f_0 &= \tilde{J}\left(6(1-6\tilde{\kappa}\chi^2)\chi^2 - 2\gamma\big|\chi(1-4\tilde{\kappa}\chi^2)\big|\right) \\ &= \tilde{J}\left(\mp 2\gamma\chi + 6\chi^2 \pm 8\gamma\tilde{\kappa}\chi^3 - 36\tilde{\kappa}\chi^4\right) \end{aligned} \tag{133}$$

with

$$\gamma = \int\limits_{[-\pi,\pi]^2} \frac{d^2k}{(2\pi)^2}\sqrt{\cos^2(k_y) + \cos^2(k_y - k_x/2) + \sin^2(k_x/2)} \approx 1.18524 \tag{134}$$

Compared to Pedros results there is also a factor 2, besides other problems, because Pedro calculates f/M $(f/2)$!

$$\frac{\partial f_0}{\partial\chi} = \tilde{J}\left(\pm 2\gamma - 12\chi\right)\left(12\tilde{\kappa}\chi^2 - 1\right) \qquad \frac{\partial^2 f_0}{\partial\chi^2} = \tilde{J}\left(12 \pm 48\gamma\tilde{\kappa}\chi - 432\tilde{\kappa}\chi^2\right) \tag{135}$$

The extremal points of the free energy are at

$$\chi' = \gamma/6, \qquad \chi'' = \pm 1/\sqrt{12\tilde{\kappa}} \tag{136}$$

$-1/\sqrt{12\tilde{\kappa}}$ is always a Maximum. $\gamma/6$ is a minimum for $\kappa \leq 6/\gamma^2$, afterwards $1/\sqrt{12\tilde{\kappa}}$ becomes a minimum:

$$f_0 = \begin{cases} \frac{\tilde{J}\gamma^2}{6}\left(\frac{\tilde{\kappa}\gamma^2}{18} - 1\right), & \tilde{\kappa} \leq 6/\gamma^2 \\ \tilde{J}\left(\frac{1}{4\tilde{\kappa}} - \frac{2\gamma}{9}\sqrt{\frac{3}{\tilde{\kappa}}}\right), & \tilde{\kappa} > 6/\gamma^2 \end{cases} \tag{137}$$

$6/\gamma^2 \approx 2.1355$. For $\kappa = 10 \Rightarrow \tilde{\kappa} = 1.\bar{6}$ and therefore $\tilde{\kappa} < 6/\gamma^2$. Hence, the free energy minimum is at

$$\chi = \gamma/6 = 0.19754, \tag{138}$$

with the free energy

$$f_0 = -1.22206\, J. \tag{139}$$

These correspond to the values I calculated using the pedestrian method above. Mind that under particle-hole transformation the calculation doesn't change (the reason basically being the absolute value in the energy, or more physically the particle-hole symmetry of the system).

0.4.3 fermi surface

The dispersion relation reads

$$\varepsilon(\vec{q}) = 2|t|\sqrt{\cos^2(q_x) + \sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right)} \tag{140}$$

I will start by calculating the fermi surface $\varepsilon(\vec{q}) = 0$, which means that the expression under the square root must vanish. Using

$$\sin^2\left(\frac{1}{2}\left(q_x + \sqrt{3}q_y\right)\right) + \cos^2\left(\frac{1}{2}\left(q_x - \sqrt{3}q_y\right)\right) = 1 + \sin(q_x)\sin\left(\sqrt{3}q_y\right) \tag{141}$$

,
I obtain the following equation for the fermi surface:

$$\underbrace{1 + \cos^2(q_0^x)}_{\in[1,2]} = \underbrace{-\sin(q_0^x)\sin\left(\sqrt{3}q_0^y\right)}_{\in[-1,1]}. \tag{142}$$

The equation can be simplified by comparing the ranges of both sides (see above)

$$\Rightarrow 1+\cos^2(q_0^x) \stackrel{!}{=} 1 \Rightarrow q_x^0 = \frac{\pi}{2} + \pi k, \quad k \in \mathbb{Z} \tag{143}$$

$$\Rightarrow \sin\left(\sqrt{3} q_0^y\right) = \begin{cases} -1 & \text{k even} \\ 1 & \text{k odd} \end{cases} \tag{144}$$

The solution (and thereby the Fermi surface) is

$$q_0^x = \frac{\pi}{2} + \pi k \tag{145}$$

$$q_0^y = \frac{1}{\sqrt{3}} \left(\frac{(-1)^{k+1}\pi}{2} + 2\pi m \right), \tag{146}$$

where $k, m \in \mathbb{Z}$. This means that the fermi surface is a collection of isolated points. More scientifically spoken: the state is a semi-metal. Two of those Fermi-points lie in the first Brillouin zone:

$$\vec{K} = \frac{\pi}{2} \begin{pmatrix} 1 \\ -1/\sqrt{3} \end{pmatrix} \quad \text{(k = m =0)} \tag{147}$$

$$\vec{K}' = \frac{\pi}{2} \begin{pmatrix} -1 \\ 1/\sqrt{3} \end{pmatrix} \quad \text{(k = -1, m =0).} \tag{148}$$

Mind you that these points are not equivalent! Evidently they are time-reversal partners.

0.4.4 small momentum expansion

To access the low-energy physics we must approximate the Hamiltonian around the fermi-points using

$$f\left(\vec{q}_0 + \delta\vec{q}\right) \approx f(\vec{q}_0) + \left(\nabla_{\vec{q}} f(\vec{q})|_{\vec{q}=\vec{q}_0}\right) \cdot \delta\vec{q}. \tag{149}$$

Around $\vec{K}$:

$$\cos\left(\vec{\delta}_1(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_1\vec{K}\right) - \sin\left(\vec{\delta}_1\vec{K}\right)\vec{\delta}_1\vec{p} = -\vec{\delta}_1\vec{p} \tag{150}$$

$$\sin\left(\vec{\delta}_2(\vec{K}+\vec{p})\right) \approx \sin\left(\vec{\delta}_2\vec{K}\right) + \cos\left(\vec{\delta}_2\vec{K}\right)\vec{\delta}_2\vec{p} = \vec{\delta}_2\vec{p} \tag{151}$$

$$\cos\left(\vec{\delta}_3(\vec{K}+\vec{p})\right) \approx \cos\left(\vec{\delta}_3\vec{K}\right) - \sin\left(\vec{\delta}_3\vec{K}\right)\vec{\delta}_3\vec{p} = -\vec{\delta}_3\vec{p} \tag{152}$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \tag{153}$$

$$\vec{\delta}_2\vec{K} = 0 \tag{154}$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \tag{155}$$

to obtain

$$\Rightarrow H_{\vec{K}} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a_{p\sigma}^\dagger & b_{p\sigma}^\dagger \end{pmatrix} \begin{pmatrix} -\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{156}$$

And around $\vec{K}' = -\vec{K}$:

$$\Rightarrow H_{\vec{K}'} \approx 2t \sum_{p,\sigma} \begin{pmatrix} a^\dagger_{p\sigma} & b^\dagger_{p\sigma} \end{pmatrix} \begin{pmatrix} \vec{\delta}_1\vec{p} & \left(\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix} \begin{pmatrix} a_{p\sigma} \\ b_{p\sigma} \end{pmatrix} \tag{157}$$

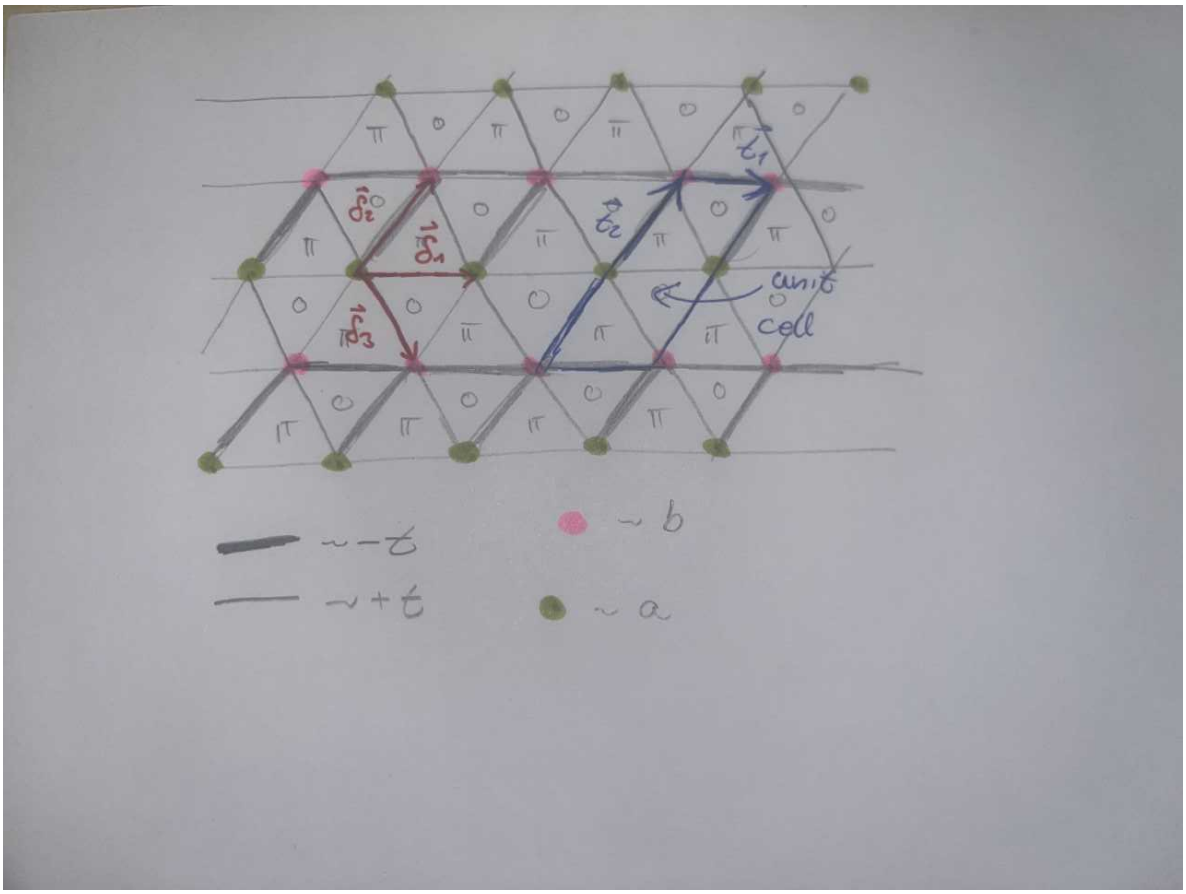
This gives the dispersion relation (for both cones)

$$\begin{aligned} \varepsilon &\approx 2t\sqrt{p_x^2 + \frac{1}{4}\left(p_x + \sqrt{3}p_y\right)^2 + \frac{1}{4}\left(p_x - \sqrt{3}p_y\right)^2} \\ &= \sqrt{6}t\sqrt{p_x^2 + p_y^2} \\ &= \sqrt{6}t|\vec{p}| \end{aligned} \tag{158}$$

Thus, the fermi velocity is $v_F = \sqrt{6}t$. This result has been reported before here [6]. $\vec{K}$ and $\vec{K}'$ are time revearsel partners, thus the two hamiltonains should map onto each other under time revearesal. I would guess that $T = -C$ in k-space, where C is complex conjugation, which would be consistent with the form of the Hamiltonian.

0.4.5 Continuum Model/ Gradient Expansion

Next I will derive the low-energy Hamilotnian in real space, as a continuum theory (/field theory). I start by writing down the real-space hamiltonian, I reshuffled term a bit compared to when I first wrote it down above, the chemical potentials are set to zero.



$$H = \frac{t}{2}\sum_{i\sigma}\left(a^\dagger_{i\sigma}a_{i+\delta_1;\sigma} - a^\dagger_{i\sigma}b_{i+\delta_2;\sigma} + a^\dagger_{i\sigma}b_{i+\delta_3;\sigma} - b^\dagger_{i\sigma}b_{i+\delta_1;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_2;\sigma} + b^\dagger_{i\sigma}a_{i+\delta_3;\sigma} + \text{h.c.}\right) \tag{159}$$

I chose a bad method for indexing the lattice position, I should eventually revise this.

The idea is to decompose the the real-space wavefuntion by its momentum - one part is the momentum is the momentum of the dirac point, the other point describes a small deviation from the dirac point (small wavevector -¿ large wavelength -¿ slowly varying in real space). This justifies a gradient expansin of the "deviation field". think of a better explenation for this. But I guess its correct in its spirit.

$$\begin{aligned} a_{i\sigma} &\sim e^{i\vec{K}\vec{R}_i}A_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}A_{K'\sigma}(\vec{R}_i) \\ b_i &\sim e^{i\vec{K}\vec{R}_i}B_{K\sigma}(\vec{R}_i) + e^{i\vec{K}'\vec{R}_i}B_{K'\sigma}(\vec{R}_i) \end{aligned} \tag{160}$$

$A_{K\sigma}, A_{K'\sigma}, B_{K\sigma}$ aqnd $B_{K'\sigma}$ are slowly varying fermionic fields. I begin with the expansion around the K-point. If the above expansion is correct, the cross terms should vanish. Show this. The protoypical method is

$$\begin{aligned} a^\dagger_{i\sigma}a_{i+\delta;\sigma} &\sim e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)A_{K\sigma}(\vec{R}_i + \vec{\delta}) \\ &\approx e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)\left(1 + \vec{\delta}\vec{\nabla}_{\vec{R}}\right)A_{K\sigma}(\vec{R}_i) \\ &= e^{i\vec{K}\vec{\delta}}A^\dagger_{K\sigma}(\vec{R}_i)\left(1 + i\vec{\delta}\vec{p}\right)A_{k\sigma}(\vec{R}_i), \end{aligned} \tag{161}$$

where $\vec{p} = -i\vec{\nabla}_{\vec{R}}$. The diagonal terms are

$$A^\dagger_{K\sigma}(\vec{R}_i)\left[e^{i\vec{K}\vec{\delta}_1}(1 + i\vec{\delta}_1\vec{p}) + e^{-i\vec{K}\vec{\delta}_1}(1 - i\vec{\delta}_1\vec{p})\right]A_{K\sigma}(\vec{R}_i) \tag{162}$$

$$B^\dagger_{K\sigma}(\vec{R}_i)\left[-e^{i\vec{K}\vec{\delta}_1}(1 + i\vec{\delta}_1\vec{p}) - e^{-i\vec{K}\vec{\delta}_1}(1 - i\vec{\delta}_1\vec{p})\right]B_{K\sigma}(\vec{R}_i). \tag{163}$$

The off-diagonal terms are

$$A^\dagger_{K\sigma}(\vec{R}_i)\frac{t}{2}\left[-e^{i\vec{\delta}_2\vec{K}}(1 + i\vec{\delta}_2\vec{p}) + e^{-i\vec{\delta}_2\vec{K}}(1 - i\vec{\delta}_2\vec{p}) + e^{i\vec{\delta}_3\vec{K}}(1 + i\vec{\delta}_3\vec{p}) + e^{-i\vec{\delta}_3\vec{K}}(1 - i\vec{\delta}_3\vec{p})\right]B_{K\sigma}(\vec{R}_i) \tag{164}$$

$$B^\dagger_{K\sigma}(\vec{R}_i)\frac{t}{2}\left[e^{i\vec{\delta}_2\vec{K}}(1 + i\vec{\delta}_2\vec{p}) - e^{-i\vec{\delta}_2\vec{K}}(1 - i\vec{\delta}_2\vec{p}) + e^{i\vec{\delta}_3\vec{K}}(1 + i\vec{\delta}_3\vec{p}) + e^{-i\vec{\delta}_3\vec{K}}(1 - i\vec{\delta}_3\vec{p})\right]A_{K\sigma}(\vec{R}_i) \tag{165}$$

Use

$$\vec{\delta}_1\vec{K} = \frac{\pi}{2} \tag{166}$$

$$\vec{\delta}_2\vec{K} = 0 \tag{167}$$

$$\vec{\delta}_3\vec{K} = \frac{\pi}{2} \tag{168}$$

to obtain

$$H_{\vec{K}} \approx \sum_\sigma 2t\int d^2\vec{R}\quad \Psi^\dagger_{K\sigma}(\vec{R})\begin{pmatrix} -\vec{\delta}_1\vec{p} & \left(-\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(-\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & \vec{\delta}_1\vec{p}\end{pmatrix}\Psi_{K\sigma}(\vec{R}), \tag{169}$$

with

$$\Psi_{K\sigma}(\vec{R}) = \begin{pmatrix} A_{K\sigma}(\vec{R}) \\ B_{K\sigma}(\vec{R}) \end{pmatrix}. \tag{170}$$

Doing the same around $\vec{K}'$ gives

$$H_{\vec{K}'} \approx \sum_{\sigma} 2t \int d^2\vec{R} \quad \Psi^{\dagger}_{K'\sigma}(\vec{R}) \begin{pmatrix} \vec{\delta}_1\vec{p} & \left(\vec{\delta}_3 - i\vec{\delta}_2\right)\vec{p} \\ \left(\vec{\delta}_3 + i\vec{\delta}_2\right)\vec{p} & -\vec{\delta}_1\vec{p} \end{pmatrix} \Psi_{K'\sigma}(\vec{R}) \tag{171}$$

Maybe it is handy to write $\vec{\delta}_1 = \vec{\delta}_2 + \vec{\delta}_3$. When Comparing this to the results obtained by expanding in reciprocal space, keep in mind that here $\vec{p}$ is an operator. So technically, one would have to do a kind of fourier transform for the fermionic fields to find the exact representation as found in recriprocal space. This is why the Hamiltonian looks so similar, even though we are still in real space here.

Imagine we did this small frequency? Fourier expansion and we found the reciprocal space Hamiltonian (replace the momentum operator by its eigenvalue!), the corresponding disperion relation would then read

$$\varepsilon(\vec{p}) = \left| 2t \begin{pmatrix} \mp\vec{\delta}_3\vec{p} \\ \vec{\delta}_2\vec{p} \\ \mp\vec{\delta}_1\vec{p} \end{pmatrix} \right| = \sqrt{6}t\sqrt{p_x^2 + p_y^2} = \sqrt{6}t|\vec{p}|, \tag{172}$$

which complies with the k-space result. In case I want to diagonalize this in operator form at some point; since the problem is non-interacting there must be a harmonic oscillator hidden in there.

0.5 0-π State on Strained Lattice

references are [4], [3], [7], [1] and [8]

0.5.1 Basics of Strain

For the purpose of introducing strain into the model, I first establish its effect on the coordinates and exchange couplings

$$\vec{R}_i \mapsto \vec{R}'_i = \vec{R}_i + \vec{U}(\vec{R}_i) \tag{173}$$

$$\vec{\delta}_{ij} \mapsto \vec{\delta}'_{ij} = \vec{\delta}_{ij} + \vec{U}(\vec{R}_j) - \vec{U}(\vec{R}_i) \tag{174}$$

$\vec{\delta}_{ij} = \vec{R}_j - \vec{R}_i$;)

$\vec{U}(\vec{R})$ is the displacement field. We approximate that $\vec{U}$ varies slowly with $\vec{R}$:

$$\vec{U}(\vec{R}_j) = \vec{U}(\vec{R}_i + \vec{\delta}_{ij}) \approx \vec{U}(\vec{R}_i) + \left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} \tag{175}$$

$$\tag{176}$$

It is convenient to redefine

$$\left(\vec{\delta}_{ij}\vec{\nabla}_{\vec{R}}\right)\vec{U}(\vec{R})\big|_{\vec{R}_i} = \underline{u}\cdot\vec{\delta}_{ij}, \tag{177}$$

where $u_{\alpha\beta} = \partial_\beta U_\alpha$ is the strain tensor.

$$\Rightarrow \vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{178}$$

Here $\underline{u}$ is taken at $\vec{R}_i$, however one can derive the same expression but with $\underline{u}$ taken at $\vec{R}_j$. In what follows I choose the expansion around $\vec{R}_i$ as shown in the main text. This however is arbitrary. Something dependening on $\vec{R}_i$ and $\vec{R}_j$ (or $(\vec{R}_i + \vec{R}_j)/2$) would seem more sensible but would complicate later expressions. I was not able to resolve this yet.

I do the same approximation for the exchange couplings

$$J(\vec{\delta}_{ij}) \mapsto J(\vec{\delta}'_{ij}) \approx J\left(\vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij}\right) \tag{179}$$

$$\approx J(\vec{\delta}_{ij}) + \left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\left(\vec{\nabla}_{\vec{\delta}} J(\vec{\delta})\big|_{\delta_{ij}}\right) \tag{180}$$

For the function $J(\vec{\delta})$ I uses the form

$$J(\vec{\delta}) = J\exp\left[-\beta\left(|\vec{\delta}|/a - 1\right)\right] \approx J\left(1 - \beta\left(|\vec{\delta}|/a - 1\right)\right), \tag{181}$$

, see e.g. [5], where a is the lattice constant and $\beta = -d\ln(J)/d\ln\left(|\vec{\delta}|\right)\big|_a$ the magneto-elastic coupling.

$$\Rightarrow \vec{\nabla}_{\vec{\delta}} J(\vec{\delta})\big|_{\delta_{ij}} \overset{|\vec{\delta}_{ij}|=a}{=} -\frac{J\beta}{a^2}\vec{\delta}_{ij}$$

$$\Rightarrow J(\vec{\delta}_{ij}) \approx J(\vec{\delta}_{ij}) - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{182}$$

Looking at the matrix elments (162 and below), one also requires the expansion of $\exp\left[i\vec{\delta}_{ij}\vec{K}\right]$. To summarize:

$$\vec{\delta}'_{ij} \approx \vec{\delta}_{ij} + \underline{u}\cdot\vec{\delta}_{ij} \tag{183}$$

$$J(\vec{\delta}'_{ij}) \approx J - \frac{J\beta}{a^2}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{184}$$

$$e^{i\vec{K}\cdot\vec{\delta}'_{ij}} \approx e^{i\vec{K}\cdot\vec{\delta}_{ij}} + ie^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right) \tag{185}$$

In this approximation the Mean field does not react to the strain at all??

0.5.2 Straining the 0-π-State

The prototypical form of the matrix elements is (expanding around $\vec{K}$)

$$J(\vec{\delta}_{ij})a^\dagger_{i\sigma}a_{i+\delta_{ij};\sigma} \sim A^\dagger_{K\sigma}\left(J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1 + i\vec{\delta}_{ij}\cdot\vec{p})\right)A_{K\sigma} \tag{186}$$

Expanding up to first order in strain I acquire the following terms

$Je^{i\vec{K}\cdot\vec{\delta}_{ij}}(1 + i\vec{\delta}_{ij}\cdot\vec{p})$	unstrained Hamiltonian/ zeroth order
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)$	contribution to $\vec{A}$ of $\mathcal{O}\left(\underline{u}^1\right)$
$-Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$-i\frac{J\beta}{a^2}e^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\left(\vec{\delta}_{ij}\cdot\vec{p}\right)$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$
$iJe^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(\underline{u}\cdot\vec{\delta}_{ij}\right)\cdot\vec{p}$	correction to v_F of $\mathcal{O}\left(\underline{u}^1\right)$

(I drop the lattice constant again from now on) I will focus on the terms of order zero + terms of either order 1 in field gradient or order 1 in strain [4]. I havent come up with a good argument for this yet. How does one even compares orders between different expansions? This gives

$$\begin{aligned} J(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}}(1+i\vec{\delta}_{ij}\cdot\vec{p}) &\sim Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\vec{p}+i\left(\vec{K}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)-\beta\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right)\right) \\ &= Je^{i\vec{K}\cdot\vec{\delta}_{ij}}\left(1+i\vec{\delta}_{ij}\cdot\left[\vec{p}+\underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right)\right]\right) \end{aligned} \tag{187}$$

Hence, on this level

$$\vec{A}=\underline{u}^T\left(\vec{K}+i\beta\vec{\delta}_{ij}\right). \tag{188}$$

This is not correct since one in has to add contributions from different local directions in the off-diagonal terms of H!

$$B=(\partial_x A_y-\partial_y A_x) \tag{189}$$

It turns out however, that $\vec{B}=0$, which is explained in what follows. The first term is simply rotationless (this was shown here [7]) and can therefore be "gauged away".

Regarding the second term; I take a step back to the dispersion relation of the unstrained model

$$H=\sum_{\vec{q}\sigma}\Psi^\dagger_{q\sigma}\cdot\left(\vec{h}(\vec{q})\cdot\vec{\sigma}\right)\cdot\Psi_{q\sigma}\quad\text{with}\quad\vec{h}(\vec{q})=2t\begin{pmatrix}\sin\left(\vec{q}\cdot\vec{\delta}_3\right)\\ \cos\left(\vec{q}\cdot\vec{\delta}_2\right)\\ \cos\left(\vec{q}\cdot\vec{\delta}_1\right)\end{pmatrix}\Rightarrow\varepsilon(\vec{q})=\left|\vec{h}(\vec{q})\right|. \tag{190}$$

By definition, $\varepsilon(\vec{K}^{(\prime)})=0$. For this to be true, every component of $\vec{h}$ must vanish individually at $\vec{K}^{(\prime)}$. This gives three conditions for $\vec{q}$, one for each direction.

Evidently each expression in $\vec{h}$ is obtained by adding contribtions from hopping forwards and backwards along $\vec{\delta}_{ij}$. However, since $\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}$ is invariant under $\vec{\delta}_{ij}\mapsto-\vec{\delta}_{ij}$ it acquires the same dispersion-like prefactor as the zeroth order, which vanishes by definition at $\vec{K}^{(\prime)}$.

So why does the term vanish? Its a mix of the special structure of the hamiltonian (i.e. the local directions decouple in the pauli-matrix basis) and the fact that the expansion term $\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}$ cant distinguish between $\vec{\delta}_{ij}$ and $-\vec{\delta}_{ij}$. The latter statement is obviously not model-specific. The same mechanism shouild be present in the π-flux square lattice. For uniform hopping (e.g. ordinary Graphene) it is absent (the decoupling of the directions is a cosnequence of the non-uniform hopping/ the mean-field).

It is not clear to me how this can be upheld under gauge transformations (wouldn't this mix directions in the Pauli-matrix basis?)

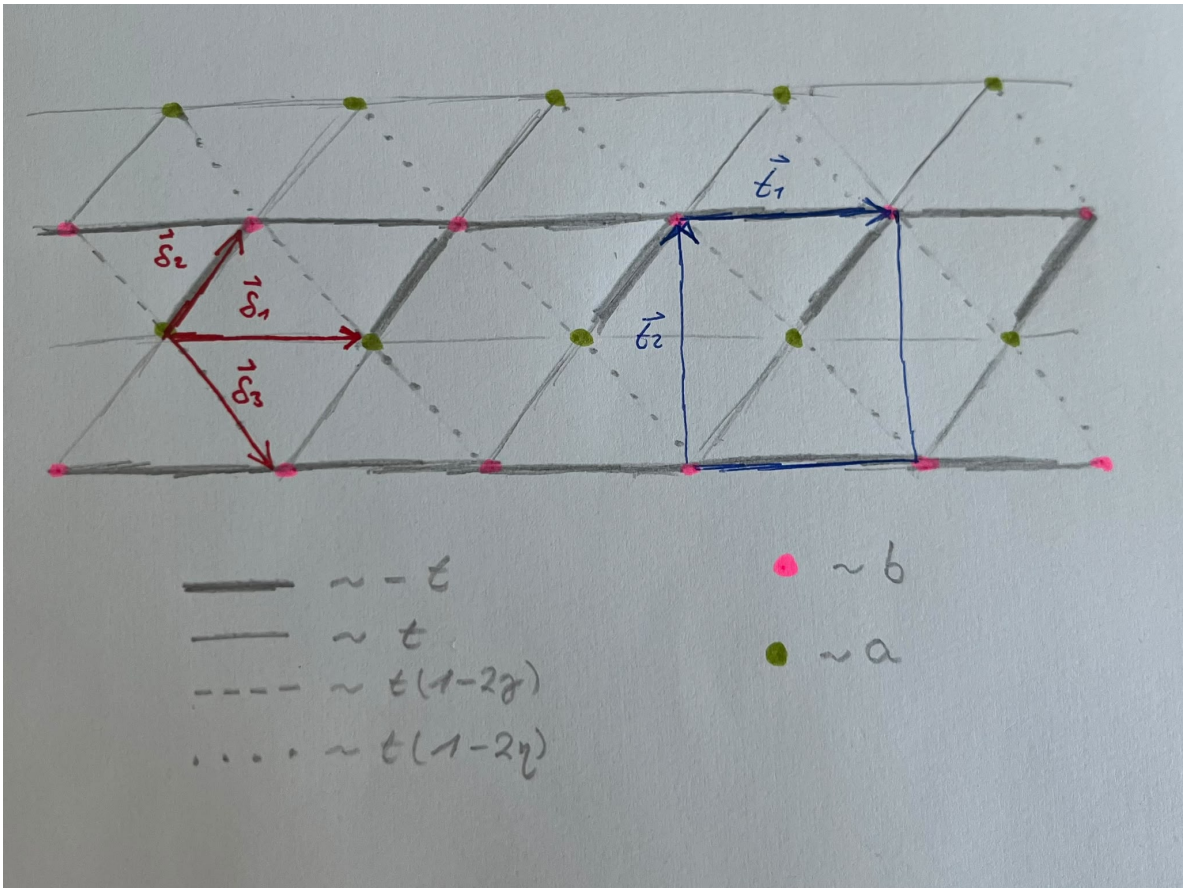
I would expect to find a pseudo-magnetic field in higher orders of strain?

0.6 Strain Engineering and other Games with the 0-π-State

This section was mostly inspired by [8].

0.6.1 Uniform Hopping

I investigate the effect of uniformly varying hopping amplitudes with one edge case reproducing a lattice topolgically equivalent to the square lattice. . I modify the hopping amplitudes in the unit cell of the two-site unit cell as follows



This gives the real space Hamiltonian

$$\begin{aligned} H = t \sum_{i\sigma} \Big(& - b^\dagger_{i\sigma} b_{i+t_1;\sigma} - b^\dagger_{i\sigma} b_{i-t_1;\sigma} + a^\dagger_{i\sigma} a_{i+t_1;\sigma} + a^\dagger_{i\sigma} a_{i-t_1;\sigma} \\ & - b^\dagger_{i\sigma} a_{i\sigma} + b^\dagger_{i\sigma} a_{i+t_2;\sigma} + (1-2\eta) b^\dagger_{i\sigma} a_{i-t_1+t_2;\sigma} + (1-2\gamma) b^\dagger_{i\sigma} a_{i+t_1} \\ & - a^\dagger_{i\sigma} b_{i\sigma} + a^\dagger_{i\sigma} b_{i-t_2;\sigma} + (1-2\eta) a^\dagger_{i\sigma} b_{i-t_1;\sigma} + (1-2\gamma) a^\dagger_{i\sigma} b_{i+t_1-t_2;\sigma} \Big). \end{aligned} \tag{191}$$

Going to reciprocal space:

$$H = t \sum_{k\sigma} \Psi^\dagger_{k\sigma} \left(\vec{h} \cdot \vec{\sigma} \right) \Psi_{k\sigma}, \tag{192}$$

with:

$$\vec{h}(\vec{q}) = 2t \begin{pmatrix} \cos(q_3) - (\eta+\gamma)\cos(q_3) \\ \sin(q_2) + (\eta-\gamma)\sin(q_3) \\ \cos(q_1) \end{pmatrix}. \tag{193}$$

Sanity check: for $\eta = \gamma = 0$ one recovers the ordinary π-flux Hmailtonian.

The (possible) Dirac points are at $\vec{h} = 0$. The σ_x suggests that there are two sets of solutions determined by:

$$\cos(q_3) = 0 \quad \text{or} \quad \eta + \gamma = 1. \tag{194}$$

For the first case use $\cos(q_1) = \cos(q_2 + q_3) = \cos(q_2)\cos(q_3) - \sin(q_2)\sin(q_3)$

$$(\sigma_z) \Rightarrow -\sin(q_2)\sin(q_3) = 0 \tag{195}$$
$$\Leftrightarrow \sin(q_2) = 0 \tag{196}$$
$$(\sigma_y) \Rightarrow (\eta - \gamma)\sin(q_3) = 0 \tag{197}$$
$$\Leftrightarrow \eta = \gamma \tag{198}$$

For the second case use $q_1 = \pm\pi/2$, (σ_y) then gives the solution for q_y (using trigonometric identities):

$$q_x = \frac{\pi}{2} \Rightarrow q_y = \frac{2}{\sqrt{3}}\arctan\left(1 - \frac{1}{\gamma}\right)$$
$$q_x = -\frac{\pi}{2} \Rightarrow q_y = -\frac{2}{\sqrt{3}}\arctan\left(1 - \frac{1}{\gamma}\right)$$

I reclassify the Dirac points by the condition on η and γ:

$\underline{\eta = \gamma :}$

$$\vec{D} = \frac{\pi}{2}\begin{pmatrix}1\\ 1/\sqrt{3}\end{pmatrix}, \quad \vec{D}' = -\frac{\pi}{2}\begin{pmatrix}1\\ 1/\sqrt{3}\end{pmatrix} \tag{199}$$

$\underline{\eta + \gamma = 1 :}$

$$\vec{D} = \begin{pmatrix}\frac{\pi}{2}\\ \frac{2}{\sqrt{3}}\arctan\left(1 - \frac{1}{\gamma}\right)\end{pmatrix}, \quad \vec{D}' = -\begin{pmatrix}\frac{\pi}{2}\\ \frac{2}{\sqrt{3}}\arctan\left(1 - \frac{1}{\gamma}\right)\end{pmatrix} \tag{200}$$

In both cases there are two Dirac points The two cases meet at $\eta = \gamma = \frac{1}{2}$, wich will be discussed in detail in the next paragraph.

$\boldsymbol{\eta = \gamma}$: Towards the square lattice

I focus on the case $\eta = \gamma$. For $\gamma = \frac{1}{2}$ one recovers a lattice of π-flux-rhombi. The idea being that this is topologcally equivalent to a π-flux-square lattice. $\gamma = 0$ just gives back the π-flux triangular lattice. It is known that in both edge cases there is no pseudomagnetic field (also these notes as well as [8]).

The Hamiltonian reads

$$\Rightarrow \vec{h}(\vec{q}) = 2t\begin{pmatrix}(1-2\gamma)\cos(q_3)\\ \sin(q_2)\\ \cos(q_1)\end{pmatrix}. \tag{201}$$

I linearize the letter around $\vec{D}$

$$\vec{h}(\vec{D} + \vec{p}) \approx \underbrace{\vec{h}(\vec{D})}_{\equiv \vec{h}^0_{\vec{D}}} + \underbrace{\left(\nabla_{\vec{q}}\vec{h}(\vec{q})\right)\cdot\vec{p}}_{\equiv \vec{h}^1_{\vec{D}}} \tag{202}$$

By definition $\vec{h}^0_{\vec{D}} = 0$. However, before making use of this fact, one still has to carry out the expansion in terms of strain. Furtheremore

$$\vec{h}^1_{\vec{D}} \equiv= 2t\begin{pmatrix}-(1-2\gamma)\sin(K_3)p_3\\ \cos(K_2)p_2\\ -\sin(K_1)p_1\end{pmatrix}. \tag{203}$$

A pseudomegnetic field would be generated by expanding $\vec{h}_0$ in strain. Expanding $\vec{h}_1$, on the other hand, only multiplies $\vec{p}$ and therefore cant give a pseudomagnetic field, at least to lowest order.

Here comes the catch; since the relevant term in the expansion of $t(\vec{\delta}_{ij})e^{i\vec{D}\cdot\vec{\delta}_{ij}}$, namely

$$-t\beta e^{i\vec{D}\cdot\vec{\delta}_{ij}}\left(\vec{\delta}_{ij}\cdot\underline{u}\cdot\vec{\delta}_{ij}\right), \tag{204}$$

are insensible to sign of the hopping direction, this reproduces the entries of $\vec{h}_{\vec{D}}$ with a factor. This then vanishes by definition. One can escape this paradigm by mixing non-parallel hopping directions in the same component of $\vec{h}(\vec{q})$, as in the next example.

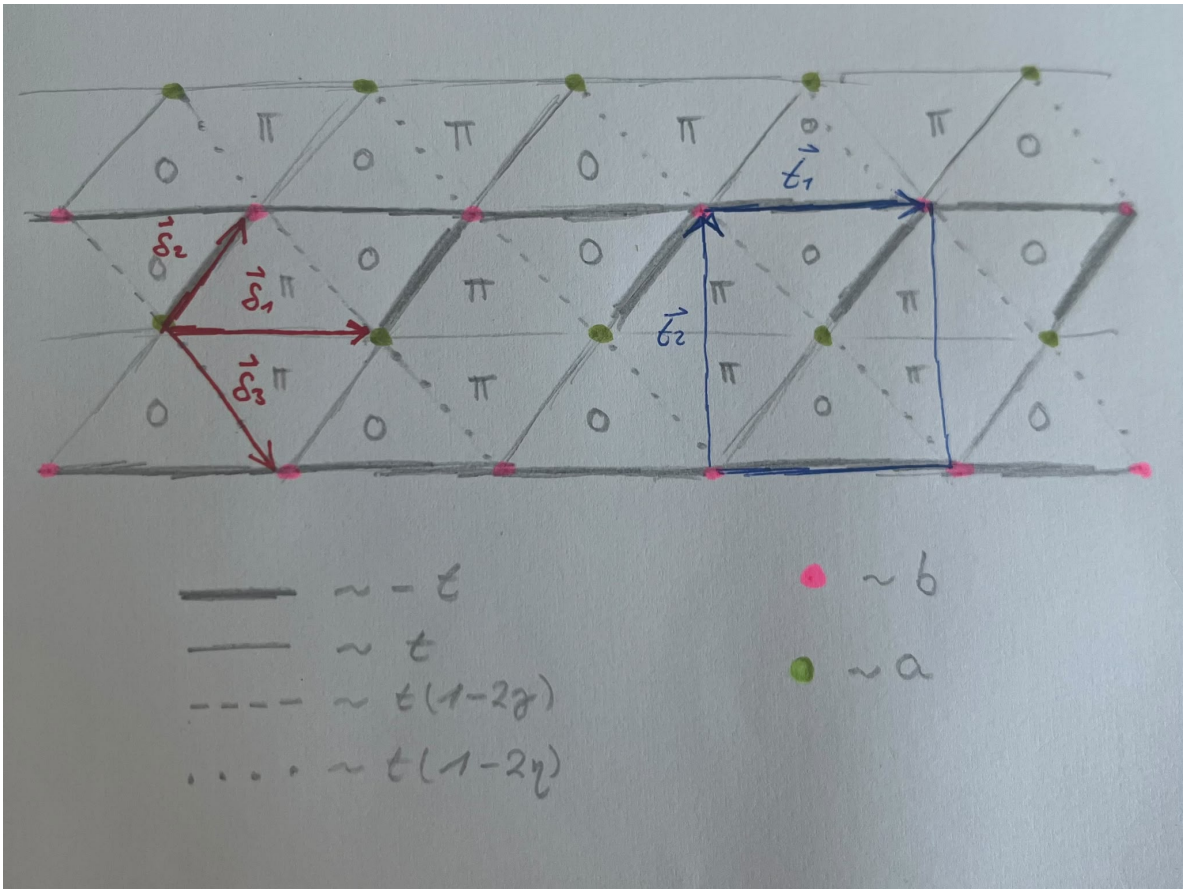
This statement only uses the form of $\vec{h}$. Shouldnt it be possible to formulate a mathematicel criterion that only depends on $\vec{h}$ that distinguishes wether the system is capable of producing pseudo-magnetic fields?

$\eta + \gamma = 1$

This choice of parameters leads to:

$$t(1-2\eta) \overset{\eta=1-\gamma}{\longrightarrow} -t(1-2\gamma). \tag{205}$$

meaning that the hoppin amplitudes along $\pm\vec{\delta}_3$, however the hopping phases alternate between 0 and π. This of course alters the flux pattern of the lattice. Acknowledging this fact it is clear that this lattice can not be transformed into the π-flux triangular lattice by choice of paramter. (This is actaully trivial, considering that $\eta = \gamma = 0$ does not satisfy $\eta + \gamma = 1$.)



The picture shows the flux pattern in the case that $\gamma < \frac{1}{2}$.

The Hamiltonian reads

$$\vec{h}(\vec{q}) = 2t \begin{pmatrix} 0 \\ \sin(q_3) + (1-2\gamma)\sin(q_2) \\ \cos(q_1) \end{pmatrix} \tag{206}$$

Coming from the discussion in the last paragraph I expect that this Hamitlonian should be able to display pseudo-magnetic fields, since the (non-parallel!) hopping-directions $\vec{q}_3$ and $\vec{q}_2$ mix in the second component. Furthermore, since in the third component only $q_1 = q_x$ appears, I expect $A_x = 0$.

Helpful identities:

$$\begin{aligned} \sin(D_2) = \cos(D_3) &= \frac{1}{\sqrt{2}} \frac{2\gamma - 1}{\sqrt{\gamma^2 + (1-\gamma)^2}}, \\ \sin(D_3) = \cos(D_2) &= \frac{1}{\sqrt{2}} \frac{1}{\sqrt{\gamma^2 + (1-\gamma)^2}} \end{aligned} \tag{207}$$

and

$$\begin{aligned} \left(\vec{\delta}_2 \cdot \underline{u} \cdot \vec{\delta}_2\right) &= \frac{1}{4}\left(u_{xx} + \sqrt{3}uxy + \sqrt{3}u_{yx} + 3u_{yy}\right) \\ \left(\vec{\delta}_3 \cdot \underline{u} \cdot \vec{\delta}_3\right) &= \frac{1}{4}\left(u_{xx} - \sqrt{3}uxy - \sqrt{3}u_{yx} + 3u_{yy}\right) \end{aligned} \tag{208}$$

I begin by linearizing the Hamiltonian:

$$h^0_{\vec{D}} = 2t \begin{pmatrix} 0 \\ \sin(D_3) + (1-2\gamma)\sin(D_2) \\ \cos(D_1) \end{pmatrix} = 0 \tag{209}$$

$$\vec{h}^1_{\vec{D}} = 2t \begin{pmatrix} 0 \\ \cos(D_3)p_3 + (1-2\gamma)\cos(D_2)p_2 \\ -\sin(D_1)p_1 \end{pmatrix} = \begin{pmatrix} 0 \\ f(\gamma)p_y \\ -p_x \end{pmatrix}, \tag{210}$$

with

$$f(\gamma) = \frac{\sqrt{3}}{\sqrt{2}} \frac{2\gamma - 1}{\sqrt{\gamma^2 + (1-\gamma)^2}}. \tag{211}$$

The next step is to reinstate the hopping directions. Starting with the σ_z components I have

$$t(\vec{\delta}_1)e^{i\vec{D}\cdot\vec{\delta}_1} + t(-\vec{\delta}_1)e^{-i\vec{D}\cdot\vec{\delta}_1}. \tag{212}$$

Expanding in strain (and only keeping the relevant term) I get

$$-t\beta\cos(D_1)\left(\vec{\delta}_1 \cdot \underline{u} \cdot \vec{\delta}_1\right) = 0, \tag{213}$$

as was expected. Doing the same for the σ_y-component I get

$$-i\left[t(\vec{\delta}_3)e^{i\vec{D}\cdot\vec{\delta}_3} - t(-\vec{\delta}_3)e^{-i\vec{D}\cdot\vec{\delta}_3}\right] + -i(1-2\gamma)\left[t(\vec{\delta}_2)e^{i\vec{D}\cdot\vec{\delta}_2} - t(-\vec{\delta}_2)e^{-i\vec{D}\cdot\vec{\delta}_2}\right] \tag{214}$$

and

$$-2t\beta\left(\sin(D_3)\left(\vec{\delta}_3 \cdot \underline{u} \cdot \vec{\delta}_3\right) + (1-2\gamma)\sin(D_2)\left(\vec{\delta}_2 \cdot \underline{u} \cdot \vec{\delta}_2\right)\right) \tag{215}$$

This term can be evaluated to

$$-f(\gamma)\frac{t\beta}{2(2\gamma-1)}\left(\frac{1-(2\gamma-1)^2}{\sqrt{3}}(u_{xx}+3u_{yy})-(1+(2\gamma-1)^2)(u_{xy}+u_{yx})\right), \tag{216}$$

which is by all means ugly but still a pseudo vectorpotential. This calculation is rather involved. I should check it using a CAS. Thi should vanish at $\eta=\gamma=\frac{1}{2}$ (or alternatively can be gauged away). Why is this not the case?

0.6.2 Staggered Hopping

Is it necessary to consider this? In the reference [8] it doesnt look like the space-dependence plays a crucial role. I guess I automatically covered this case with the unifrom hopping stuff.

Staggered hoppig in the way that it was introduced in [8] manifestly doesnt work in the thermodynamic limit since there must be a x after which (coming from 0) the sign of the hopping (and with it the flux pattern) flips.

On the other hand one could interpret this as the linear approximation of the strain-effect on t. So the real question would be which strain pattern produces this. It is however, questionable if such a pattern physcially exist (judging from the form of the modified t). Does this mean this approach is more honest than uniformly chanigng the hopping amplitudes? Of course, there the argmument is that such a lattice can in principle exist.

0.6.3 On-Site Potential

References

[1] A. H. Castro Neto et al. "The electronic properties of graphene". In: *Rev. Mod. Phys.* 81 (1 Jan. 2009), pp. 109–162. DOI: 10.1103/RevModPhys.81.109. URL: https://link.aps.org/doi/10.1103/RevModPhys.81.109.

[2] Tom Drechsler and Matthias Vojta. "Emergent Chiral Metal near a Kondo Breakdown Quantum Phase Transition". In: *Phys. Rev. Lett.* 134 (10 Mar. 2025), p. 106503. DOI: 10.1103/PhysRevLett.134.106503. URL: https://link.aps.org/doi/10.1103/PhysRevLett.134.106503.

[3] Alexander L. Kitt et al. "Lattice-corrected strain-induced vector potentials in graphene". In: *Phys. Rev. B* 85 (11 Mar. 2012), p. 115432. DOI: 10.1103/PhysRevB.85.115432. URL: https://link.aps.org/doi/10.1103/PhysRevB.85.115432.

[4] Mary Madelynn Nayga, Stephan Rachel, and Matthias Vojta. "Magnon Landau Levels and Emergent Supersymmetry in Strained Antiferromagnets". In: *Phys. Rev. Lett.* 123 (20 Nov. 2019), p. 207204. DOI: 10.1103/PhysRevLett.123.207204. URL: https://link.aps.org/doi/10.1103/PhysRevLett.123.207204.

[5] Vitor M. Pereira, A. H. Castro Neto, and N. M. R. Peres. "Tight-binding approach to uniaxial strain in graphene". In: *Phys. Rev. B* 80 (4 July 2009), p. 045401. DOI: 10.1103/PhysRevB.80.045401. URL: https://link.aps.org/doi/10.1103/PhysRevB.80.045401.

[6] Stephan Rachel et al. "Quantum Paramagnet in a π Flux Triangular Lattice Hubbard Model". In: *Phys. Rev. Lett.* 114 (16 Apr. 2015), p. 167201. DOI: 10.1103/PhysRevLett.114.167201. URL: https://link.aps.org/doi/10.1103/PhysRevLett.114.167201.

[7] M. Ramezani Masir, D. Moldovan, and F.M. Peeters. "Pseudo magnetic field in strained graphene: Revisited". In: *Solid State Communications* 175-176 (Dec. 2013), pp. 76–82. ISSN: 0038-1098. DOI: 10.1016/j.ssc.2013.04.001. URL: http://dx.doi.org/10.1016/j.ssc.2013.04.001.

[8] Junsong Sun et al. “Pseudomagnetic fields in square lattices”. In: *Phys. Rev. B* 108 (20 Nov. 2023), p. 205149. DOI: 10.1103/PhysRevB.108.205149. URL: https://link.aps.org/doi/10.1103/PhysRevB.108.205149.